Digit Recognition
Using Perceptron Training Procedure

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Abstract

We have tried to make use of a set of perceptrons that can recognize digits shown on a 7-segment display. Each perceptron is trained by perceptron convergence procedure (PCP), to recognize one of 10 digits from 0 to 9. We have interfaced an LCD screen to display outputs.

1 Introduction

Hardware implementations of neural networks are theoretically more efficient than their software counterparts. With this motivation, we attempted to implement an elementary system comprising a set of perceptrons. Using the perceptron convergence procedure, we trained it to identify digits 0 to 9, encoded in the form of a seven-segment display.

2 Block Diagram (top-level)

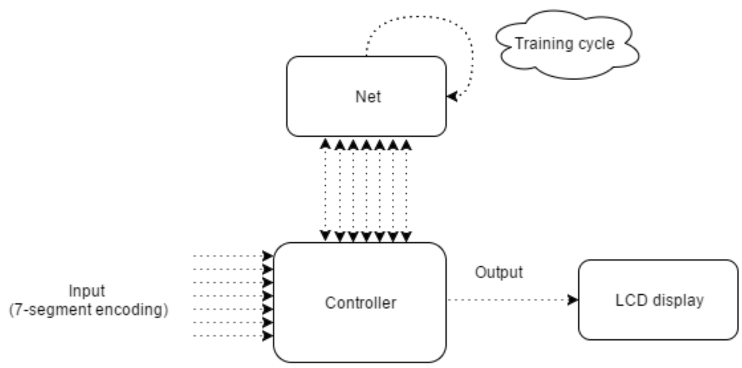

Figure 1: Block Diagram

Our project can be described by the adjoining block diagram depicting the inputs and output-interface. The controller entity serves as the top-level entity. It communicates the input to the neural net, which then proceeds into a training cycle. Upon convergence, the LCD displays a message on the screen asking the user to input a digit for recognition.

3 Working Principle

We made a set of 10 perceptrons, each to be trained for one of the digits (0,1,..,9). In the training phase, each perceptron was provided a known input combination, corresponding to its digit. This is followed by the training phase, wherein each perceptron iterates till it reaches the decision surface in the weight space. This is followed by the test phase, where all the perceptrons are tested with the currently trained weights. Each perceptron is successively subjected through train-test cycles, until all the weights are trained.

3.1 Schematic

The circuit schematic, for a single perceptron looks as follows:

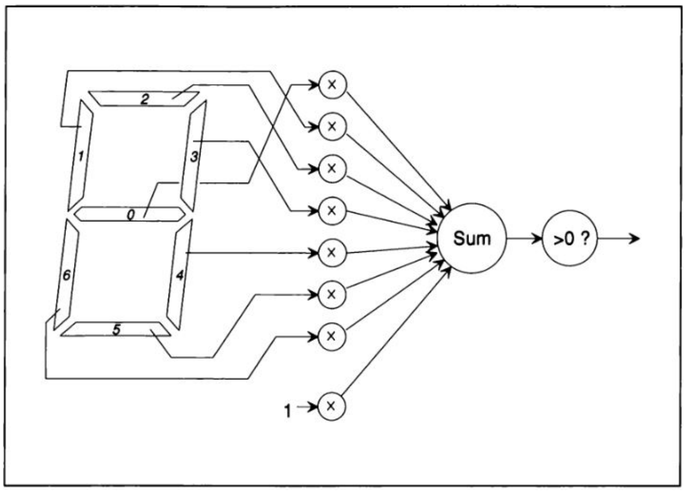

Figure 2: Perceptron schematic

Our system, that recognizes 10 different digits has 10 such perceptrons, each with 7 external inputs (and 1 threshold input, fixed by the internal logic). The interconnection between the input pins and perceptrons are depicted graphically in the following figure:

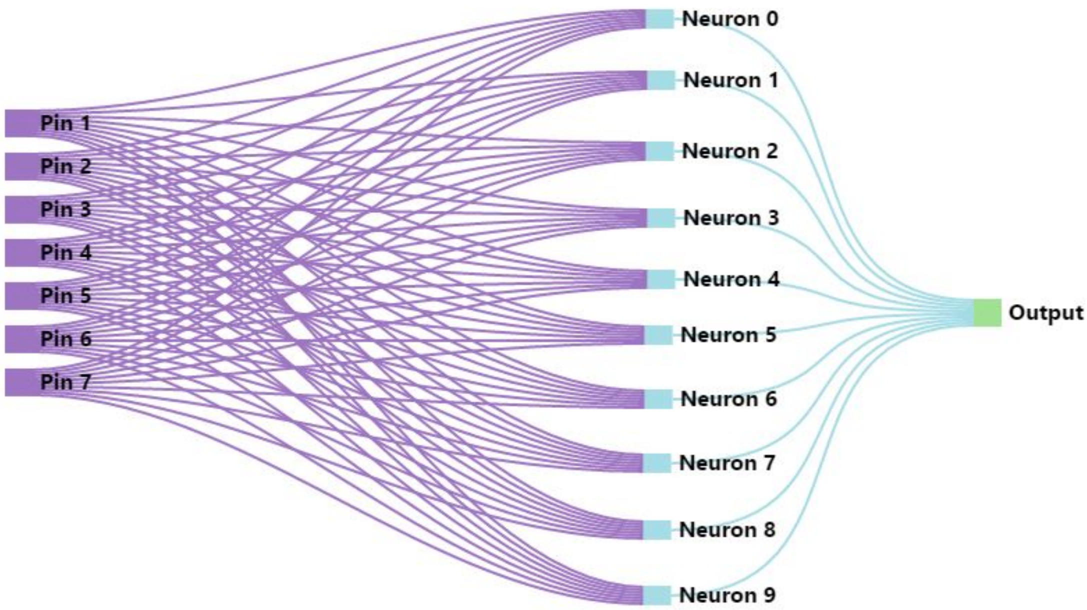

Figure 3: Structure of the Net

3.2 Algorithm

The training code used follows the following algorithm, for a single perceptron at a time.

1. When in training mode
   (a) if the training output is greater than the expected output, then the input vector is subtracted from the weight vector

(b) otherwise the input vector is added to the weight vector

2. Test the trained perceptron weights, by providing corresponding inputs for the digits.
3. Accept input from user and display the digit corresponding to the perceptron that fires with respect to the given input data.

Once training of each perceptron is completed, the system prompts the user to input a digit, again with 7-segment encoding. Once input into the controller in-ports, only one of the perceptrons should fire, identifying exactly which digit the input is.

4 I/O interfacing

Input

Input is provided, currently, from a set of seven switches which model the seven segment LED display.
Alternatively, we had planned to display the digits as required for the training phase on an 8x8 LED matrix, modified by Arduino code to behave like a large 7-segment display. Seven photodiodes were planned to be used to sense which of the seven segments are lit up. The output of this photodiode array serves as our input to the perceptrons.

Output

The output from the code, at various stages, is displayed on a 16x2 LCD display. This includes prompting the user to display particular digits in the training phase, asking for input from the user, after training is over, and displaying the final output.

5 LCD interfacing : details

We are using a 16 x 2 character LCD display for output in this project. For this purpose we designed an interface to make use of the LCD as and when required by the rest of the circuit. The structure is depicted in the figure below

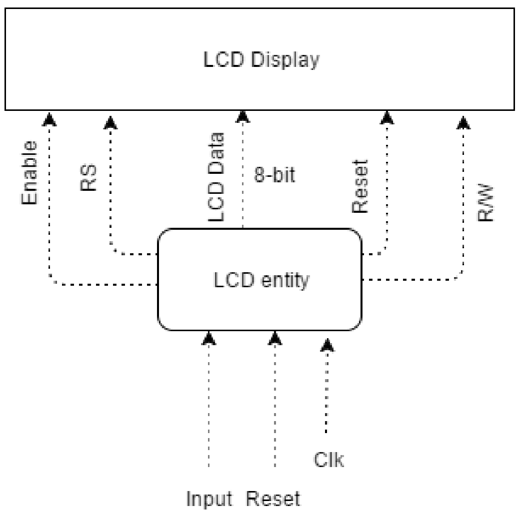

Figure 4: LCD block diagram

6 Work Distribution

1. Basic perceptron convergence: Aditya
2. Design and initial implementation of neural net: Aditya
3. Program interacting FSM to interface input and neural net: Aditya
4. LCD interfacing: Arkya
5. LED matrix: Arkya

7 Testing

Input all the digits (7-segment encoded) for training. Provide the user input when prompted on the LCD screen, by means of a set of switches. Output should identify which digit was entered by user.